

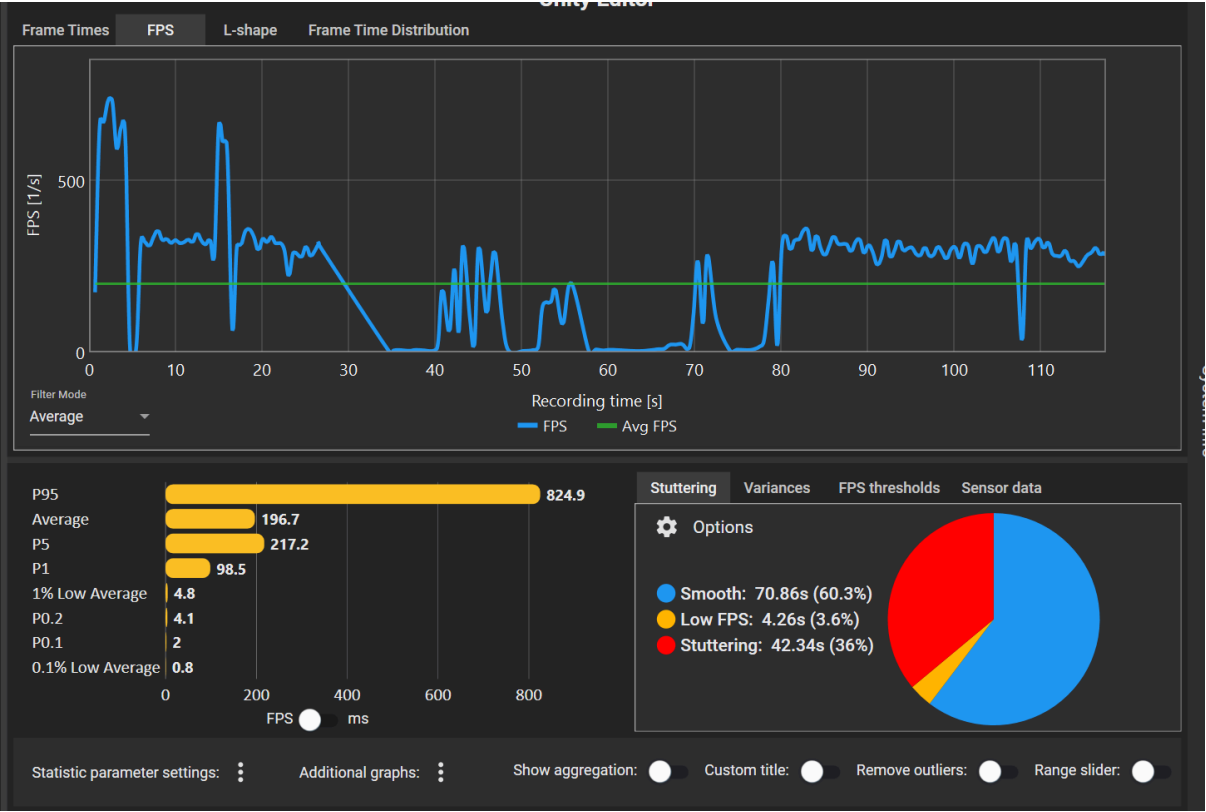
Benchmark report / Comparaison avec & sans post-processing / Profiler data

- configuration 1:

PC portable gamer

Composant	Spécification
Modèle	MSI Pulse 16 AI (C1V)
Processeur	Intel Core Ultra 9 185H – 16 cœurs / 22 threads, jusqu'à 5,1 GHz
Carte graphique	NVIDIA GeForce RTX 4060 Laptop – 8 Go GDDR6
Accélérateur IA (NPU)	Intel AI Boost – accélération matérielle pour l'inférence locale
Mémoire vive	96 Go DDR5 – double canal (2 × 48 Go)
Systèmes d'exploitation	Windows 11 / Linux (dual boot) + WSL2 pour le développement
Capacité du disque	SSD M2 2to

Indicateur	Valeur Mesurée
FPS Moyen	355fps
Utilisation CPU Moyenne	24%
Consommation RAM Max	1.56go
Consommation VRAM (GPU)	2.02go
Outils de benchmark	CapFrameX

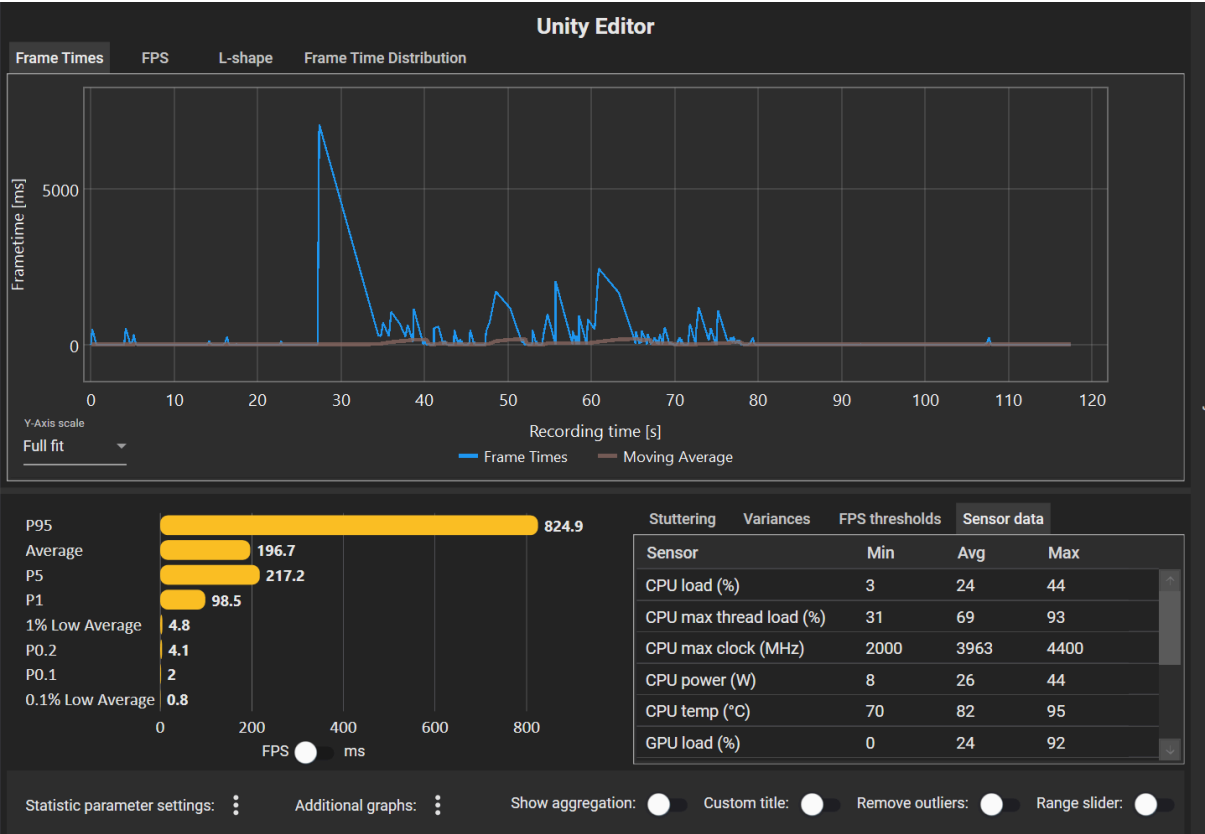


Sensor items			Sensor statistics for selected record		
Sensor	Type	Use for logging	Sensor	Min	Avg
Core #1 E	Load	☐	GPU Core (MHz)	210	1268
Core #2 E	Load	☐	GPU Power Limit	0	0
Core #3 E	Load	☐	GPU Core (%)	0	24
Core #4 E	Load	☐	GPU Limit Time (%)	0	0
Core #5 E	Load	☐	GPU Power (W)	7.1	18.7
Core #6 E	Load	☐	GPU Core (°C)	62	66
Core #7 E	Load	☐	GPU Memory Dedicated (GB)	1.94	2.02
Core #8 E	Load	☐	CPU Max (MHz)	2000	3966
Core #9 P Thread #1	Load	☐	CPU Max (%)	31	69
Core #9 P Thread #2	Load	☐	CPU Total (%)	3	24
Core #10 P Thread #1	Load	☐	CPU Package (W)	8	26
Core #10 P Thread #2	Load	☐	CPU Package (°C)	70	82
Core #11 P Thread #1	Load	☐	RAM Game Used (GB)	1.36	1.56
Core #11 P Thread #2	Load	☐			
Core #12 P Thread #1	Load	☐			
Core #12 P Thread #2	Load	☐			
Core #13 P Thread #1	Load	☐			

☒ Log sensors 250

Group control

Evaluate multiple entries Evaluation Aggregate

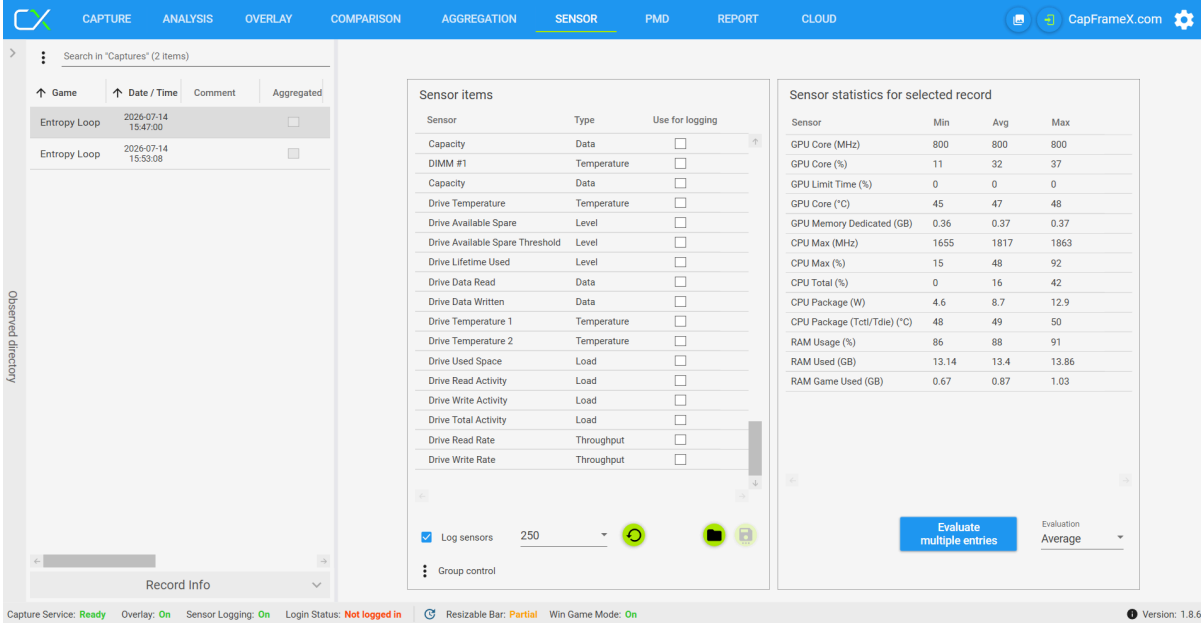
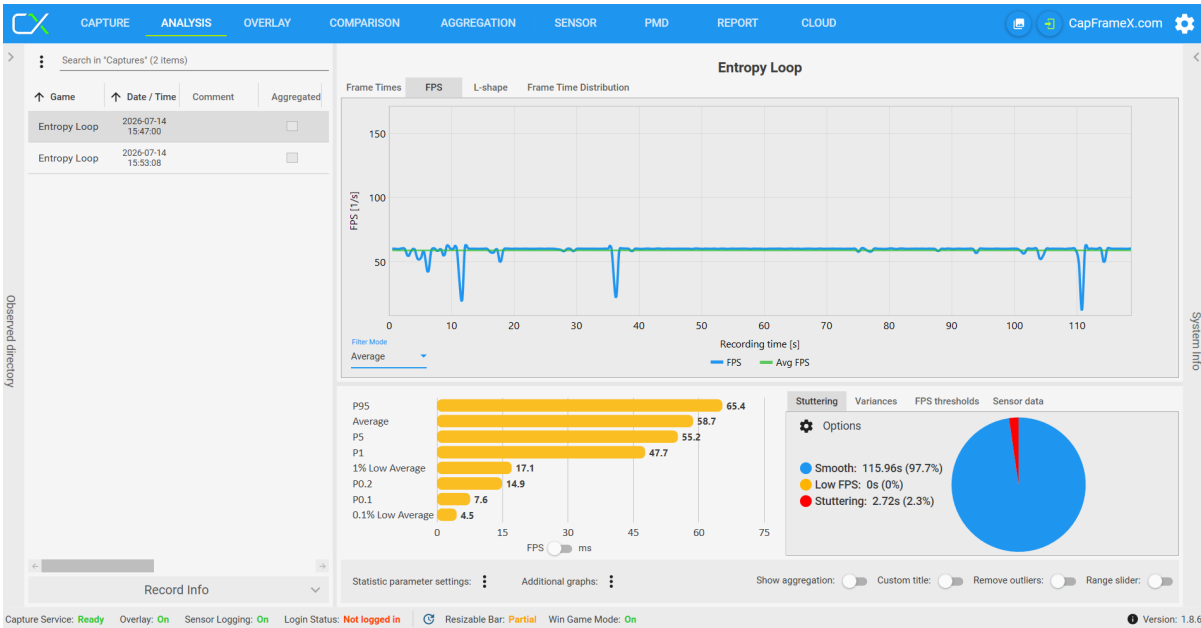


- configuration 2:

PC portable

Composant	Spécification
Modèle	Dell Inspiron 16 5645
Processeur	AMD Ryzen 7 8840U Hawk Point (3.3 GHz, 8 cœurs, TDP 20-30W)
Carte graphique	AMD Radeon 780M intégrée au processeur
Mémoire vive	16 Go (2 x 8 Go) DDR5 5600 MHz
Systèmes d'exploitation	Windows 11 64 bits / Parrot (dual boot)
Capacité du disque	SSD M.2 de 1 To (NVMe PCIe)

Indicateur	Valeur Mesurée
FPS Moyen	58.7 FPS
Utilisation CPU Moyenne	16%
Consommation RAM Max	1.03Go
Consommation VRAM (GPU)	0.37Go (383 Mo)
Outils de benchmark	CapFrameX

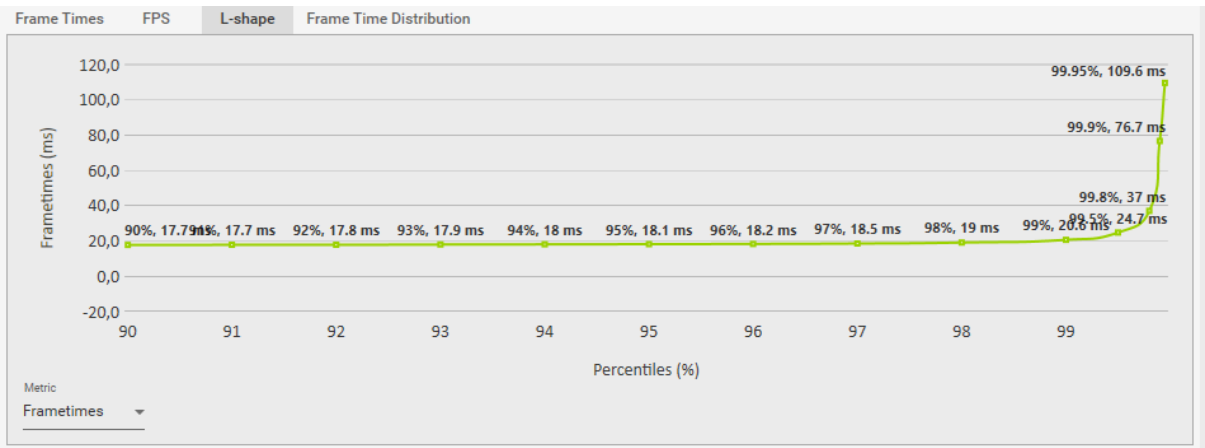
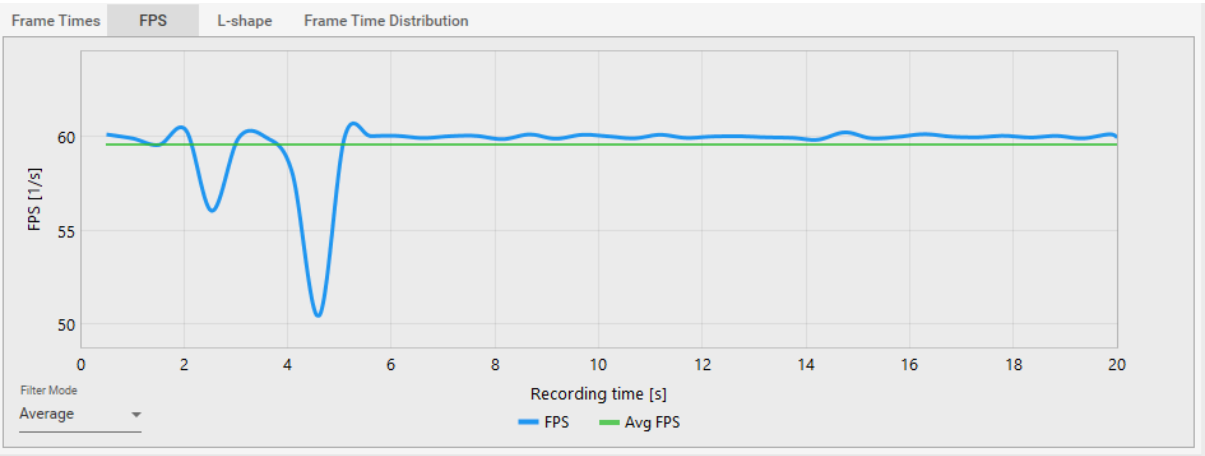
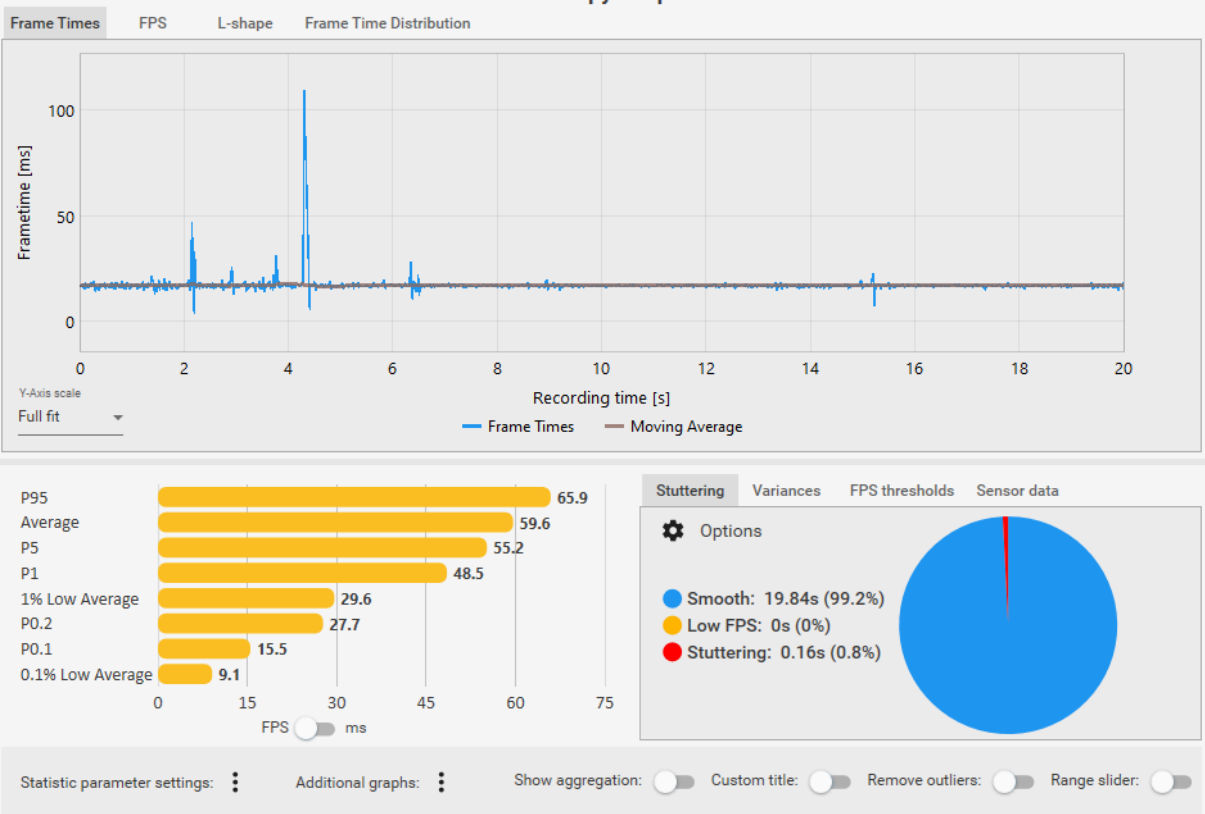


- configuration 3:

PC fixe

Composant	Spécification
Modèle	Sur mesure
Processeur	AMD Ryzen 5 3600
Carte graphique	NVIDIA GeForce RTX 4060
Mémoire vive	16GB (2x8GB) 3200MT/s
Systèmes d'exploitation	windows 10
Capacité du disque	2To

Indicateur	Valeur Mesurée
FPS Moyen	60
Utilisation CPU Moyenne	~15%
Consommation RAM Max	580m0
Consommation VRAM (GPU)	1.5go
Outils de benchmark	CapFrameX



- configuration 4:

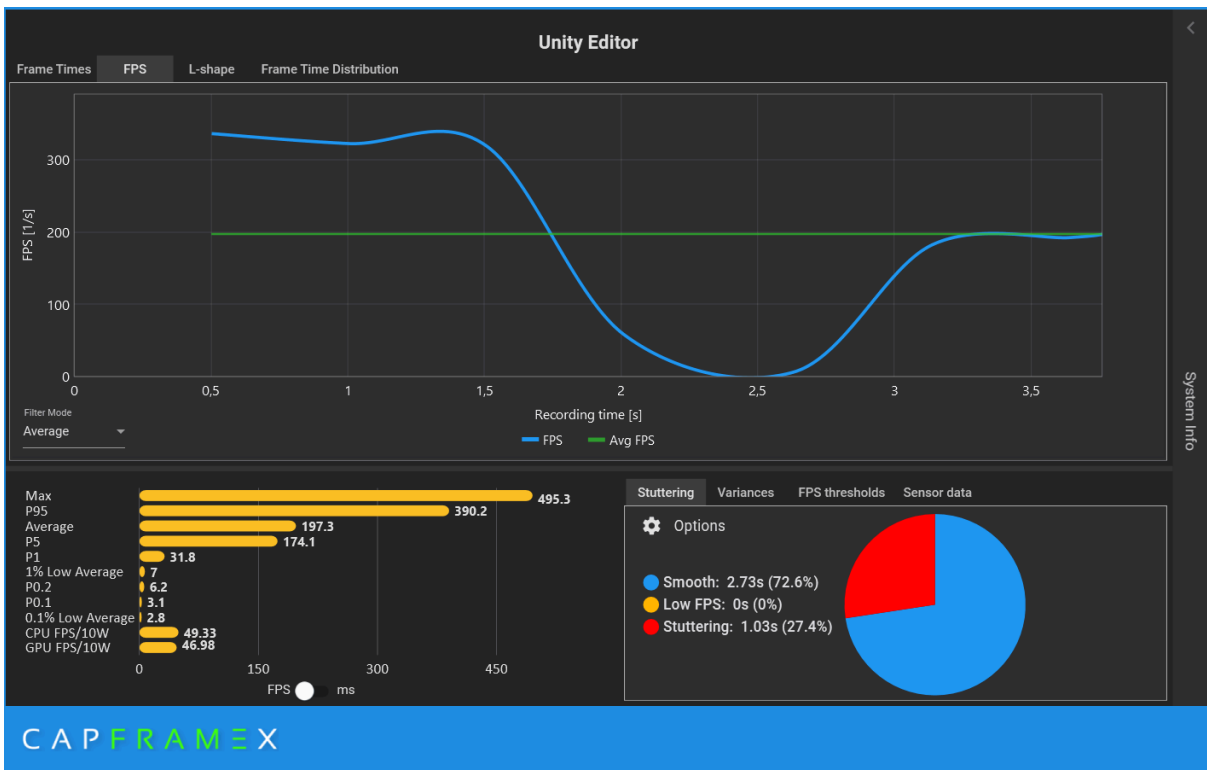
PC gamer

Composant	Spécification
Modèle	Dell G15 5510
Processeur	Intel(R) Core(TM) i7-10870H CPU @ 2.20GHz, 2208 MHz, 8 cœur(s), 16 processeur(s) logique(s)
Carte graphique	NVIDIA GeForce RTX 3060 - 6Go GDDR6
Mémoire vive	16 Go
Systèmes d'exploitation	Windows 11
Capacité du disque	512 Go

Indicateur	Valeur Mesurée
FPS Moyen	160.2 FPS
Utilisation CPU Moyenne	~11%
Consommation RAM Max	~280 Mo
Consommation VRAM (GPU)	23%
Outils de benchmark	CapFrameX

Test n°1: Chargement du niveau (CPU)

Ce test a été effectué en utilisant uniquement le CPU.



Sensor items

Sensor	Type	Use for logging
Core #1	Temperature	☐
Core #2	Temperature	☐
Core #3	Temperature	☐
Core #4	Temperature	☐
Core #5	Temperature	☐
Core #6	Temperature	☐
Core #7	Temperature	☐
Core #8	Temperature	☐
CPU Package	Temperature	☒
Core #1 Distance to TjMax	Temperature	☐
Core #2 Distance to TjMax	Temperature	☐
Core #3 Distance to TjMax	Temperature	☐
Core #4 Distance to TjMax	Temperature	☐
Core #5 Distance to TjMax	Temperature	☐
Core #6 Distance to TjMax	Temperature	☐
Core #7 Distance to TjMax	Temperature	☐
Core #8 Distance to TjMax	Temperature	☐
Core #1	Clock	☐

☒ Log sensors 250

Sensor statistics for selected record

Sensor	Min	Avg	Max
GPU Core (MHz)	1260	1787	1950
GPU Power Limit	0	0	0
GPU Core (%)	0	33	53
GPU Limit Time (%)	0	0	0
GPU Power (W)	28.7	41.9	50.9
GPU Core (°C)	64	67	69
GPU Memory Dedicated Game (GB)	0	0	0
CPU Max (MHz)	4000	4057	4200
CPU Effective (MHz)	555	944	1339
CPU Max Effective (MHz)	1653	2824	4136
CPU Max (%)	46	70	93
CPU Total (%)	11	23	36
CPU Package (W)	27.9	40.2	47.5
CPU Memory (W)	1.1	2	2.4
CPU Package (°C)	81	86	89
RAM Usage (%)	81	82	83
Virtual RAM Usage (%)	60	61	61

Evaluate multiple entries

Evaluation Aggregate

CAPFRAMEX

Sensor items

Sensor	Type	Use for logging
Core #1	Temperature	☐
Core #2	Temperature	☐
Core #3	Temperature	☐
Core #4	Temperature	☐
Core #5	Temperature	☐
Core #6	Temperature	☐
Core #7	Temperature	☐
Core #8	Temperature	☐
CPU Package	Temperature	☒
Core #1 Distance to TjMax	Temperature	☐
Core #2 Distance to TjMax	Temperature	☐
Core #3 Distance to TjMax	Temperature	☐
Core #4 Distance to TjMax	Temperature	☐
Core #5 Distance to TjMax	Temperature	☐
Core #6 Distance to TjMax	Temperature	☐
Core #7 Distance to TjMax	Temperature	☐
Core #8 Distance to TjMax	Temperature	☐
Core #1	Clock	☐

☒ Log sensors 250   

Sensor statistics for selected record

Sensor	Min	Avg	Max
GPU Core (°C)	64	67	69
GPU Memory Dedicated Game (GB)	0	0	0
CPU Max (MHz)	4000	4057	4200
CPU Effective (MHz)	555	944	1339
CPU Max Effective (MHz)	1653	2824	4136
CPU Max (%)	46	70	93
CPU Total (%)	11	23	36
CPU Package (W)	27.9	40.2	47.5
CPU Memory (W)	1.1	2	2.4
CPU Package (°C)	81	86	89
RAM Usage (%)	81	82	83
Virtual RAM Usage (%)	60	61	61
RAM Used (GB)	12.82	12.94	13.07
RAM Available (GB)	2.71	2.84	2.96
RAM Game Used (GB)	1.36	1.48	1.62
Virtual RAM Used (GB)	19.02	19.19	19.37
Virtual RAM Available (GB)	12.19	12.37	12.54

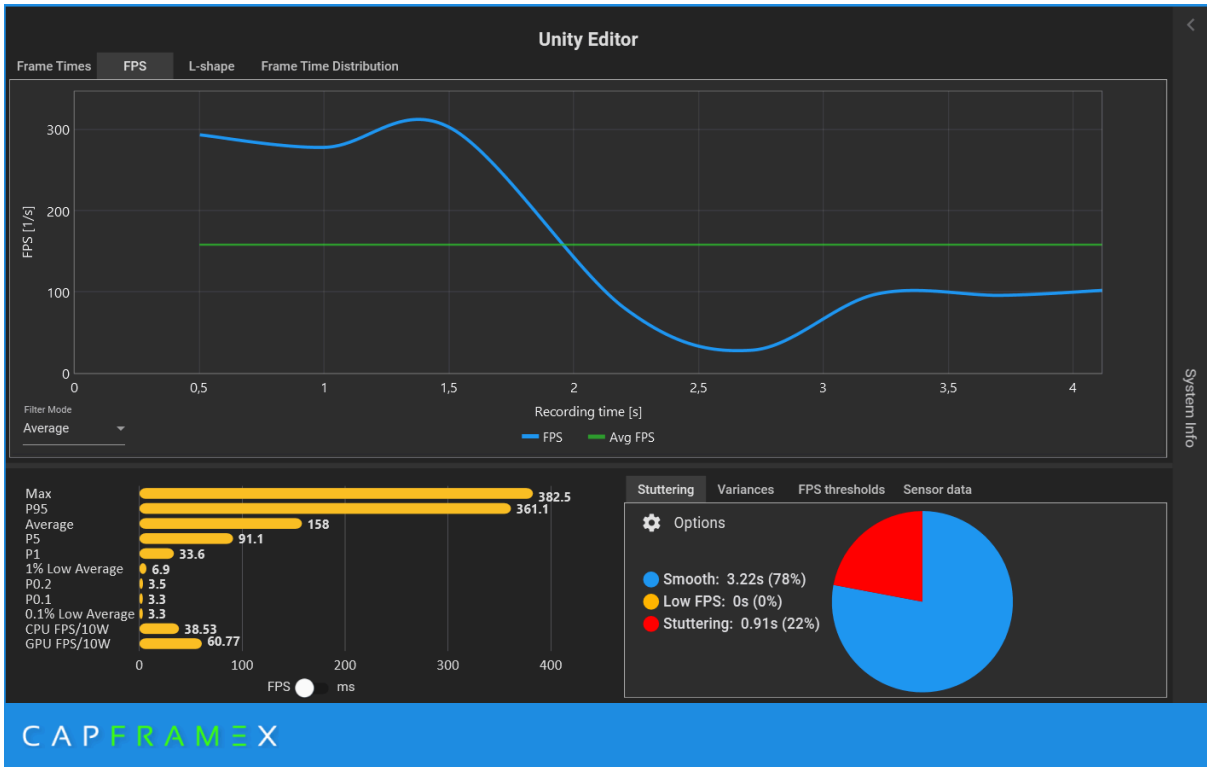
Evaluate multiple entries

Evaluation Aggregate

CAPFRAME X

Test 1.bis : Chargement du niveau (GPU)

Ce test a été fait en utilisant le GPU principalement



Sensor items

Sensor	Type	Use for logging
Core #1 Thread #1	Load	☐
Core #1 Thread #2	Load	☐
Core #2 Thread #1	Load	☐
Core #2 Thread #2	Load	☐
Core #3 Thread #1	Load	☐
Core #3 Thread #2	Load	☐
Core #4 Thread #1	Load	☐
Core #4 Thread #2	Load	☐
Core #5 Thread #1	Load	☐
Core #5 Thread #2	Load	☐
Core #6 Thread #1	Load	☐
Core #6 Thread #2	Load	☐
Core #7 Thread #1	Load	☐
Core #7 Thread #2	Load	☐
Core #8 Thread #1	Load	☐
Core #8 Thread #2	Load	☐
CPU Total	Load	☒
CPU Max	Load	☒

Log sensors 250

Sensor statistics for selected record

Sensor	Min	Avg	Max
GPU Core (MHz)	720	1064	1230
GPU Power Limit	0	0	0
GPU Core (%)	0	29	60
GPU Limit Time (%)	0	0	0
GPU Power (W)	23.2	26.3	28.9
GPU Core (°C)	67	68	68
GPU Memory Dedicated Game (GB)	0	0	0
CPU Max (MHz)	4000	4043	4200
CPU Effective (MHz)	671	939	1227
CPU Max Effective (MHz)	1656	2522	2944
CPU Max (%)	48	65	79
CPU Total (%)	12	24	31
CPU Package (W)	32.9	40.5	46.7
CPU Memory (W)	1.8	2.2	2.7
CPU Package (°C)	83	87	92
RAM Usage (%)	92	92	92
Virtual RAM Usage (%)	91	91	91

Evaluate multiple entries

Evaluation Aggregate

CAPFRAMEX

Sensor items

Sensor	Type	Use for logging
Core #1 Thread #1	Load	☐
Core #1 Thread #2	Load	☐
Core #2 Thread #1	Load	☐
Core #2 Thread #2	Load	☐
Core #3 Thread #1	Load	☐
Core #3 Thread #2	Load	☐
Core #4 Thread #1	Load	☐
Core #4 Thread #2	Load	☐
Core #5 Thread #1	Load	☐
Core #5 Thread #2	Load	☐
Core #6 Thread #1	Load	☐
Core #6 Thread #2	Load	☐
Core #7 Thread #1	Load	☐
Core #7 Thread #2	Load	☐
Core #8 Thread #1	Load	☐
Core #8 Thread #2	Load	☐
CPU Total	Load	☒
CPU Max	Load	☒

☒ Log sensors
 250

Sensor statistics for selected record

Sensor	Min	Avg	Max
GPU Core (°C)	67	68	68
GPU Memory Dedicated Game (GB)	0	0	0
CPU Max (MHz)	4000	4043	4200
CPU Effective (MHz)	671	939	1227
CPU Max Effective (MHz)	1656	2522	2944
CPU Max (%)	48	65	79
CPU Total (%)	12	24	31
CPU Package (W)	32.9	40.5	46.7
CPU Memory (W)	1.8	2.2	2.7
CPU Package (°C)	83	87	92
RAM Usage (%)	92	92	92
Virtual RAM Usage (%)	91	91	91
RAM Used (GB)	14.48	14.5	14.52
RAM Available (GB)	1.26	1.28	1.3
RAM Game Used (GB)	1.84	1.85	1.85
Virtual RAM Used (GB)	29.68	29.72	29.75
Virtual RAM Available (GB)	2.83	2.86	2.9

Evaluate multiple entries

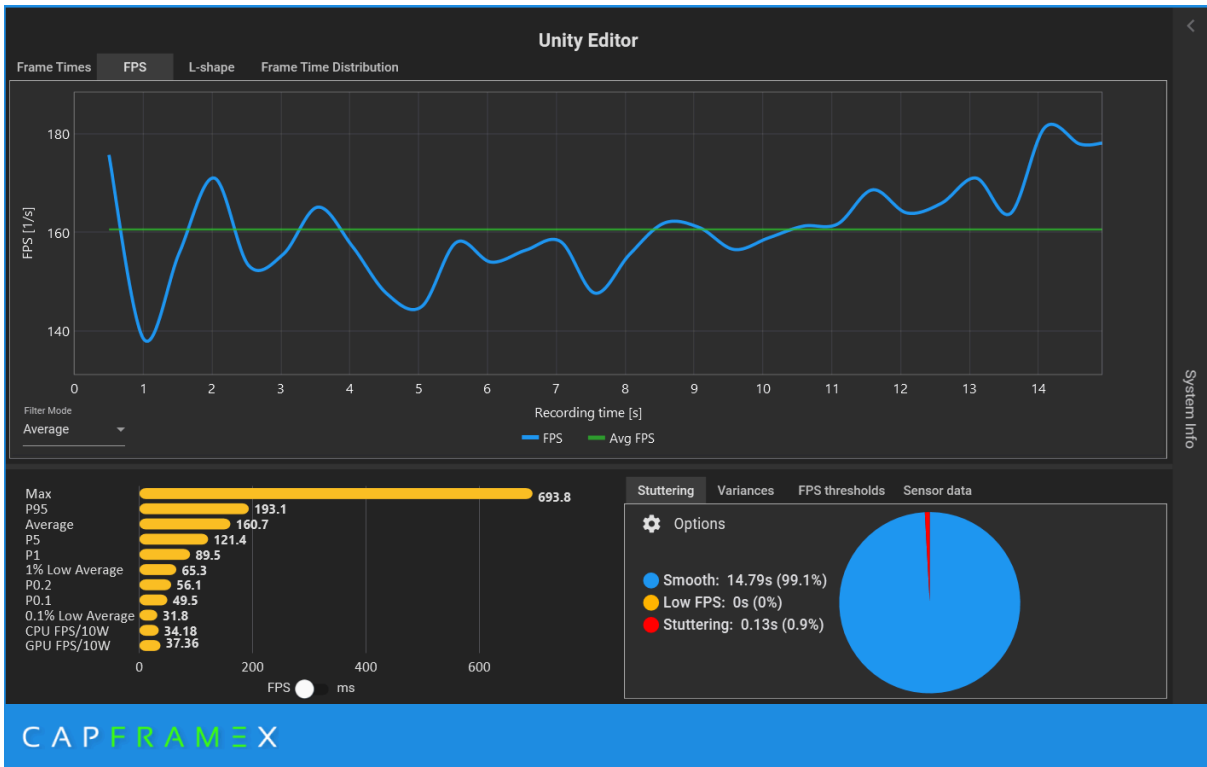
Evaluation Aggregate

CAPFRAMEX

Avec l'aide du GPU, la génération de la carte d'un niveau se fait plus rapidement, une différence de 0.3 seconde de moins qu'avec le CPU.

De plus lors du chargement on gagne ~30FPS avec le GPU qu'avec le CPU, on peut constater ceci surtout au moment ou on rentre dans le niveau (time code 2.5 sec sur chaque images)

Test n°2: Gameplay d'un niveau (défaite)



Sensor items

Sensor	Type	Use for logging
Core #1	Temperature	☐
Core #2	Temperature	☐
Core #3	Temperature	☐
Core #4	Temperature	☐
Core #5	Temperature	☐
Core #6	Temperature	☐
Core #7	Temperature	☐
Core #8	Temperature	☐
CPU Package	Temperature	☒
Core #1 Distance to TjMax	Temperature	☐
Core #2 Distance to TjMax	Temperature	☐
Core #3 Distance to TjMax	Temperature	☐
Core #4 Distance to TjMax	Temperature	☐
Core #5 Distance to TjMax	Temperature	☐
Core #6 Distance to TjMax	Temperature	☐
Core #7 Distance to TjMax	Temperature	☐
Core #8 Distance to TjMax	Temperature	☐
Core #1	Clock	☐

☒ Log sensors 250

Sensor statistics for selected record

Sensor	Min	Avg	Max
GPU Core (MHz)	1387	1533	1897
GPU Power Limit	0	0	0
GPU Core (%)	26	38	42
GPU Limit Time (%)	0	0	0
GPU Power (W)	35.4	42.7	63.6
GPU Core (°C)	70	71	76
GPU Memory Dedicated Game (GB)	0	0	0
CPU Max (MHz)	4000	4000	4000
CPU Effective (MHz)	1093	1329	1701
CPU Max Effective (MHz)	2300	2844	3425
CPU Max (%)	57	70	84
CPU Total (%)	27	32	45
CPU Package (W)	40	47	56.1
CPU Memory (W)	2.1	2.3	2.6
CPU Package (°C)	86	91	94
RAM Usage (%)	82	83	83
Virtual RAM Usage (%)	61	61	62

Evaluate multiple entries Evaluation: Aggregate

CAPFRAMEX

Sensor items

Sensor	Type	Use for logging
Core #1	Temperature	☐
Core #2	Temperature	☐
Core #3	Temperature	☐
Core #4	Temperature	☐
Core #5	Temperature	☐
Core #6	Temperature	☐
Core #7	Temperature	☐
Core #8	Temperature	☐
CPU Package	Temperature	☒
Core #1 Distance to TjMax	Temperature	☐
Core #2 Distance to TjMax	Temperature	☐
Core #3 Distance to TjMax	Temperature	☐
Core #4 Distance to TjMax	Temperature	☐
Core #5 Distance to TjMax	Temperature	☐
Core #6 Distance to TjMax	Temperature	☐
Core #7 Distance to TjMax	Temperature	☐
Core #8 Distance to TjMax	Temperature	☐
Core #1	Clock	☐

☒ Log sensors 250   

Sensor statistics for selected record

Sensor	Min	Avg	Max
GPU Core (°C)	70	71	76
GPU Memory Dedicated Game (GB)	0	0	0
CPU Max (MHz)	4000	4000	4000
CPU Effective (MHz)	1093	1329	1701
CPU Max Effective (MHz)	2300	2844	3425
CPU Max (%)	57	70	84
CPU Total (%)	27	32	45
CPU Package (W)	40	47	56.1
CPU Memory (W)	2.1	2.3	2.6
CPU Package (°C)	86	91	94
RAM Usage (%)	82	83	83
Virtual RAM Usage (%)	61	61	62
RAM Used (GB)	12.99	13.02	13.06
RAM Available (GB)	2.72	2.76	2.79
RAM Game Used (GB)	1.63	1.65	1.66
Virtual RAM Used (GB)	19.33	19.37	19.43
Virtual RAM Available (GB)	12.13	12.19	12.23

<

Evaluate multiple entries

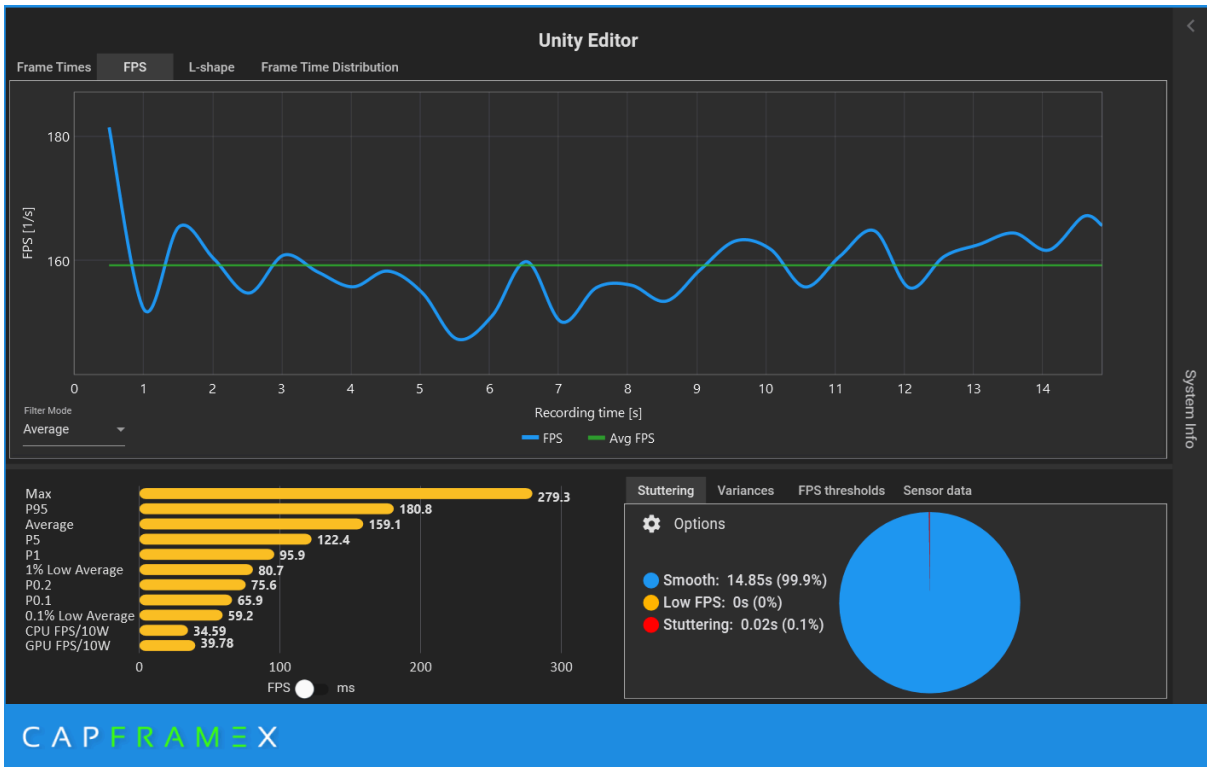
>

Evaluation

Aggregate

CAPFRAMEX

Test n°3: Gameplay d'un niveau (réussite)



Sensor items

Sensor	Type	Use for logging
Core #1	Temperature	☐
Core #2	Temperature	☐
Core #3	Temperature	☐
Core #4	Temperature	☐
Core #5	Temperature	☐
Core #6	Temperature	☐
Core #7	Temperature	☐
Core #8	Temperature	☐
CPU Package	Temperature	☒
Core #1 Distance to TjMax	Temperature	☐
Core #2 Distance to TjMax	Temperature	☐
Core #3 Distance to TjMax	Temperature	☐
Core #4 Distance to TjMax	Temperature	☐
Core #5 Distance to TjMax	Temperature	☐
Core #6 Distance to TjMax	Temperature	☐
Core #7 Distance to TjMax	Temperature	☐
Core #8 Distance to TjMax	Temperature	☐
Core #1	Clock	☐

Log sensors 250

Sensor statistics for selected record

Sensor	Min	Avg	Max
GPU Core (MHz)	1327	1471	1777
GPU Power Limit	0	0	0
GPU Core (%)	32	38	42
GPU Limit Time (%)	0	0	0
GPU Power (W)	36.3	40.2	49.7
GPU Core (°C)	71	72	75
GPU Memory Dedicated Game (GB)	0	0	0
CPU Max (MHz)	4000	4004	4200
CPU Effective (MHz)	1193	1296	1503
CPU Max Effective (MHz)	2373	2881	3363
CPU Max (%)	60	72	87
CPU Total (%)	25	31	35
CPU Package (W)	43.3	45.9	49.7
CPU Memory (W)	2.2	2.3	2.5
CPU Package (°C)	88	92	96
RAM Usage (%)	83	83	83
Virtual RAM Usage (%)	61	62	62

Evaluate multiple entries

Evaluation: Aggregate

CAPFRAMEX

Sensor items

Sensor	Type	Use for logging
Core #1	Temperature	☐
Core #2	Temperature	☐
Core #3	Temperature	☐
Core #4	Temperature	☐
Core #5	Temperature	☐
Core #6	Temperature	☐
Core #7	Temperature	☐
Core #8	Temperature	☐
CPU Package	Temperature	☒
Core #1 Distance to TjMax	Temperature	☐
Core #2 Distance to TjMax	Temperature	☐
Core #3 Distance to TjMax	Temperature	☐
Core #4 Distance to TjMax	Temperature	☐
Core #5 Distance to TjMax	Temperature	☐
Core #6 Distance to TjMax	Temperature	☐
Core #7 Distance to TjMax	Temperature	☐
Core #8 Distance to TjMax	Temperature	☐
Core #1	Clock	☐

☒ Log sensors 250   

Sensor statistics for selected record

Sensor	Min	Avg	Max
GPU Core (°C)	71	72	75
GPU Memory Dedicated Game (GB)	0	0	0
CPU Max (MHz)	4000	4004	4200
CPU Effective (MHz)	1193	1296	1503
CPU Max Effective (MHz)	2373	2881	3363
CPU Max (%)	60	72	87
CPU Total (%)	25	31	35
CPU Package (W)	43.3	45.9	49.7
CPU Memory (W)	2.2	2.3	2.5
CPU Package (°C)	88	92	96
RAM Usage (%)	83	83	83
Virtual RAM Usage (%)	61	62	62
RAM Used (GB)	13.06	13.07	13.08
RAM Available (GB)	2.7	2.71	2.72
RAM Game Used (GB)	1.66	1.67	1.67
Virtual RAM Used (GB)	19.4	19.41	19.42
Virtual RAM Available (GB)	12.14	12.15	12.16

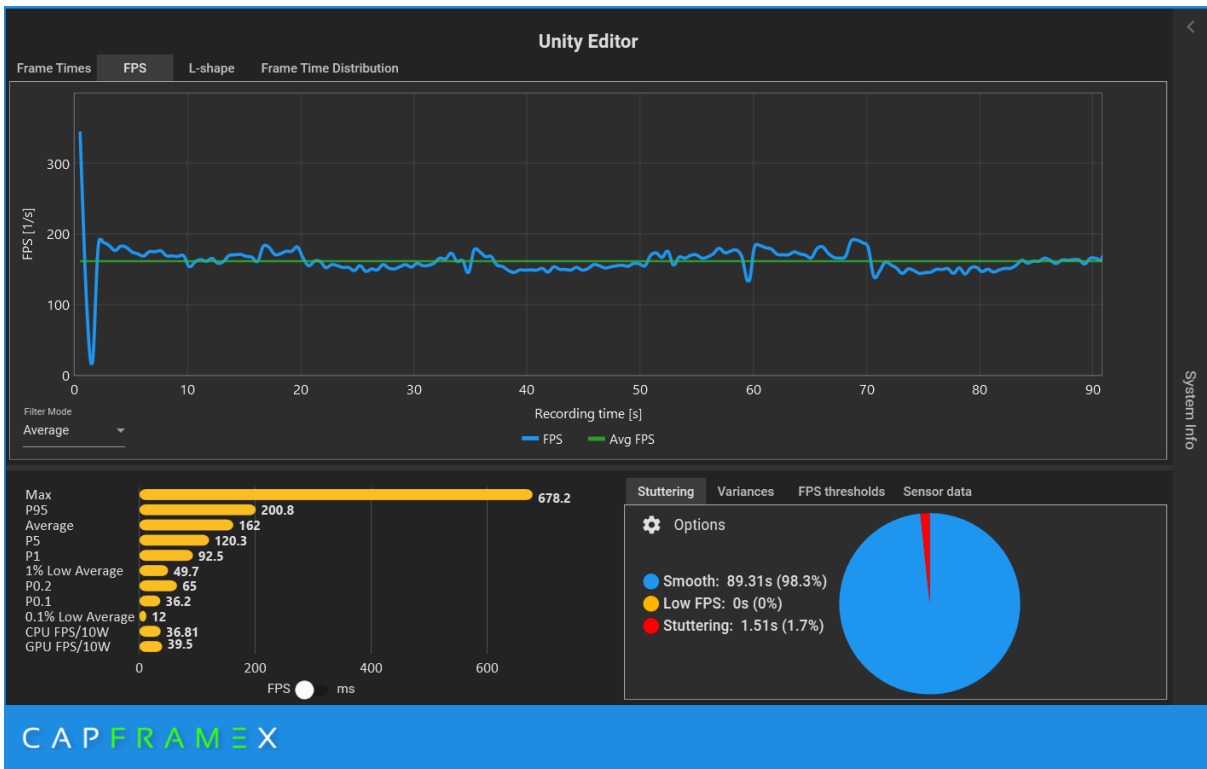
 

Evaluate multiple entries

Evaluation Aggregate

CAPFRAMEX

Test n°4: Du début jusqu'à la fin d'un niveau



Sensor items

Sensor	Type	Use for logging
Core #1	Temperature	☐
Core #2	Temperature	☐
Core #3	Temperature	☐
Core #4	Temperature	☐
Core #5	Temperature	☐
Core #6	Temperature	☐
Core #7	Temperature	☐
Core #8	Temperature	☐
CPU Package	Temperature	☒
Core #1 Distance to TjMax	Temperature	☐
Core #2 Distance to TjMax	Temperature	☐
Core #3 Distance to TjMax	Temperature	☐
Core #4 Distance to TjMax	Temperature	☐
Core #5 Distance to TjMax	Temperature	☐
Core #6 Distance to TjMax	Temperature	☐
Core #7 Distance to TjMax	Temperature	☐
Core #8 Distance to TjMax	Temperature	☐
Core #1	Clock	☐

☒ Log sensors 250

Sensor statistics for selected record

Sensor	Min	Avg	Max
GPU Core (MHz)	1252	1517	1852
GPU Power Limit	0	0	0
GPU Core (%)	1	38	45
GPU Limit Time (%)	0	0	0
GPU Power (W)	31.7	41.4	56.7
GPU Core (°C)	68	74	79
GPU Memory Dedicated Game (GB)	0	0	0
CPU Max (MHz)	3900	3998	4200
CPU Effective (MHz)	456	1272	1783
CPU Max Effective (MHz)	1685	2988	4368
CPU Max (%)	55	75	90
CPU Total (%)	13	31	41
CPU Package (W)	26	44.4	52.7
CPU Memory (W)	1.2	2.3	2.5
CPU Package (°C)	82	92	96
RAM Usage (%)	83	83	84
Virtual RAM Usage (%)	61	62	62

Evaluate multiple entries

Evaluation Aggregate

CAPFRAMEX

Sensor items

Sensor	Type	Use for logging
Core #1	Temperature	☐
Core #2	Temperature	☐
Core #3	Temperature	☐
Core #4	Temperature	☐
Core #5	Temperature	☐
Core #6	Temperature	☐
Core #7	Temperature	☐
Core #8	Temperature	☐
CPU Package	Temperature	☒
Core #1 Distance to TjMax	Temperature	☐
Core #2 Distance to TjMax	Temperature	☐
Core #3 Distance to TjMax	Temperature	☐
Core #4 Distance to TjMax	Temperature	☐
Core #5 Distance to TjMax	Temperature	☐
Core #6 Distance to TjMax	Temperature	☐
Core #7 Distance to TjMax	Temperature	☐
Core #8 Distance to TjMax	Temperature	☐
Core #1	Clock	☐

☒ Log sensors 250   

Sensor statistics for selected record

Sensor	Min	Avg	Max
GPU Core (°C)	68	74	79
GPU Memory Dedicated Game (GB)	0	0	0
CPU Max (MHz)	3900	3998	4200
CPU Effective (MHz)	456	1272	1783
CPU Max Effective (MHz)	1685	2988	4368
CPU Max (%)	55	75	90
CPU Total (%)	13	31	41
CPU Package (W)	26	44.4	52.7
CPU Memory (W)	1.2	2.3	2.5
CPU Package (°C)	82	92	96
RAM Usage (%)	83	83	84
Virtual RAM Usage (%)	61	62	62
RAM Used (GB)	13.07	13.13	13.2
RAM Available (GB)	2.58	2.65	2.71
RAM Game Used (GB)	1.64	1.74	1.79
Virtual RAM Used (GB)	19.34	19.45	19.52
Virtual RAM Available (GB)	12.04	12.11	12.22

Evaluate multiple entries

Evaluation Aggregate

CAPFRAMEX

- configuration 5:

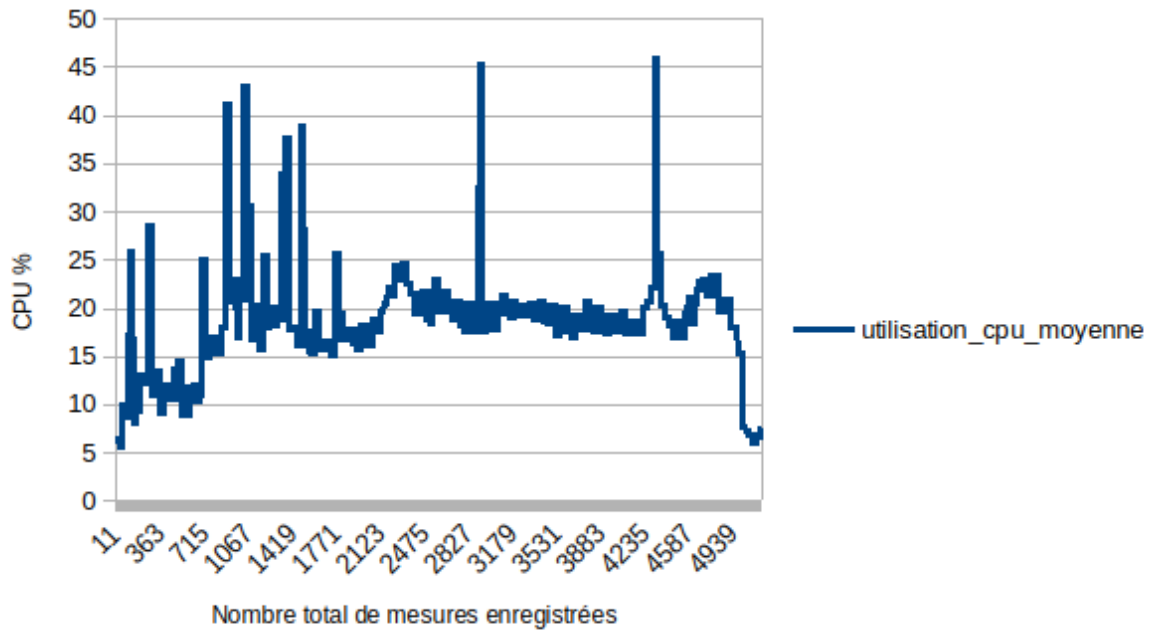
PC portable

Composant	Spécification
Modèle	DELL vostro
Processeur	11th Gen Intel Core i5-11320H × 8
Carte graphique	Iris Xe Graphics
Mémoire vive	16Go
Systèmes d'exploitation	Ubuntu 24.04.4 LTS
Capacité du disque	500Go

Indicateur	Valeur Mesurée
FPS Moyen	60.1 FPS
Utilisation CPU Moyenne	~18.8 %
Consommation RAM Max	~560 Mo
Consommation VRAM (GPU)	0 %
Outils de benchmark	Lutris avec MangoHud et Wine, Unity

Données récupérées sur la version 0.1.5 du jeu.

Les pics sont les génération de map et les combats avec beaucoup d'entités.
Graphique réalisé avec les données provenant du .csv généré par MangoHud :



Données provenant du .csv généré par MangoHud sur la version 0.1.7 du jeu :

0.1% Min FPS	1% Min FPS	97% Percentile FPS	Average FPS	GPU Load	CPU Load
5.3	26.9	64.9	59.8	0.0	23.3

Benchmark réalisé en utilisant unity :



Benchmark réalisé avant optimisation de map :



Il y a une moyenne d'environ 30 FPS avant l'optimisation de la map.

Il y a 60 FPS lorsque l'on est dans le menu du jeu.

Les pics que l'on peut remarquer, apparaissent lors d'une génération de map.

Configuration minimal requise :

Composant	Spécification
Processeur	Intel Core i3 or equivalent
Carte graphique	Intel HD Graphics or equivalent
Mémoire vive	4Go
Systèmes d'exploitation	Window / linux (avec wine ou équivalent)
Espace de stockage requis	1Go

Configuration recommandé requise :

Composant	Spécification
Processeur	Intel Core i5 or equivalent
Carte graphique	Intel HD Graphics or equivalent
Mémoire vive	8Go
Systèmes d'exploitation	Window / linux (avec wine ou équivalent)
Espace de stockage requis	2Go